Data Representation

Outline

- Number Representation
  - Decimal, Binary(二进制), and Hexadecimal(十六进制)
  - Bit(比特, 二进制位) and Byte(字节)
  - Conversion(转换) Between Number Systems
  - Negative Integers Representation
  - Basic Arithmetic(算术) Operations with Binary(二进制) Numbers
- Text Representation
  - ASCII Code and Unicode
- Data Compression(压缩)

Analogue (模拟) VS. Digital (数字)

- **Analogue(模拟) Data**: A continuous(连续) representation (表示) which precisely describes the actual information.



- **Digital data (数字)**: A discrete(离散的) representation which breaks the information into units and approximates (近似) the actual information



- In computers, **digital data** is used

Data Representation

- Data come in many forms
  - Numbers: 233, 11.04, -29, …
  - Text: “hello, world!” “汪！ ”
  - Audio: .mp3, .wav
  - Picture: .bmp, .gif
  - Video: .avi, .mp4

This training

Other major courses (□ □ □)

- Data must be represented in a way that
  - Captures (抓取) the essence (本质) of the information
  - is convenient for computer to process
- All data are stored in computers as binary(二进制) digits

Natural Numbers

- Natural numbers(自然数)
  - Zero and any number obtained by repeatedly adding one to it
- Negative Numbers(负数)
  - A value less than 0, with a – sign
- Positive Numbers(正数)
  - A value bigger than 0
- Integers(整数)
  - A natural number, a negative number

Decimal Number System

- A human usually has four fingers and a thumb on each hand, giving a total of ten digits over both hands
- 10 digits:
  - 0,1,2,3,4,5,6,7,8,9
- Also called base-10 number system,
  - Or Hindu-Arabic, or Arabic system



- Counting in base-10
  - 1,2,…,9,10,11,…,19,20,21,…,99,100,…
- Decimal number in expanded notation
  - $234 = 2*100 + 3*10 + 4*1 = 2*10^2 + 3*10^1 + 4*10^0$

Binary(二进制) Number System

- Binary(二进制) number system has only two digits
  - 0, 1
  - Also called base-2 system

- Counting in binary(二进制) system
  - 0, 1, 10, 11, 100, 101, 110, 111, 1000,….

- Binary(二进制) number in expanded notation
  - $(1011)_2 = 1*2^3 + 0*2^2 + 1*2^1 + 1*2^0$
  - $(1011)_2 = 1*8 + 0*4 + 1*2 + 1*1 = (11)_{10}$

Hexadecimal(十六进制)

- Hexadecimal(十六进制) number system has 16 digits
  - 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F
  - Also called base-16 system
- Counting in Hexadecimal(十六进制)
  - 0,1,…,F,10,11,…,1F,20,…FF,100,…
- Hexadecimal(十六进制) number in expanded notation
  - $(FF)_{16} = 15*16^1 + 15*16^0 = (255)_{10}$

Some Numbers to Remember

Power of 2
$2^{-2} = \frac{1}{4} = 0.25$
$2^{-1} = \frac{1}{2} = 0.5$
$2^0 = 1$
$2^1 = 2$
$2^2 = 4$
$2^3 = 8$
$2^4 = 16$
$2^5 = 32$
$2^6 = 64$
$2^7 = 128$
$2^8 = 256$
$2^9 = 512$
$2^{10} = 1{,}024$
$2^{14} = 16{,}384$
$2^{15} = 32{,}768$

Decimal	4-Bit Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Positional Numbering System

- The value of a digit in a number depends on
  - the digit itself and
  - the position of the digit within the number
- 1 in 123 is different from 1 in 321
  - 123: 1 hundred
  - 321: 1 unit

Why Binary(二进制)?

- A computer is a Binary(二进制) machine
- It knows only ones and zeroes
- Easy to implement in electronic circuits(电子线路)
- Reliable
- Cheap

Bit(比特, 二进制位) and Byte(字节)

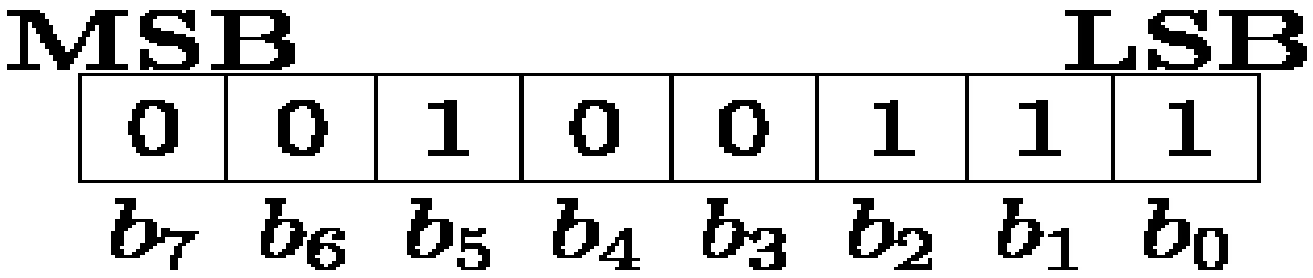
- BIT (比特，二进制位) = Binary(二进制) digIT, "0" or "1"
- State of on or off (high or low) of a computer circuit
- Kilo 1K = 2^{10} = 1024 ≈ 10^3
- Mega 1M = 2^{20} = 1,048,576 ≈ 10^6
- Giga 1G = 2^{30} = 1,073,741,824 ≈ 10^9

Bit(比特) and Byte(字节)

- Byte(字节) is the basic unit of addressable(可寻址的) memory (内存)
- 1 Byte(字节) = 8 Bits(二进制位)
- The right-most bit is called the LSB

  Least Significant Bit
- The Left-most bit is called the MSB

  Most Significant Bit

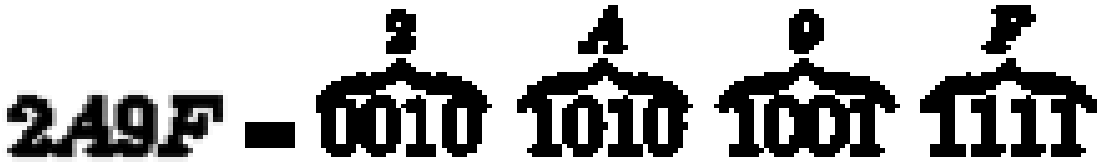


Why Hexadecimal(十六进制)?

- When dealing with large values, binary(二进制) numbers quickly become too lengthy.
- Hexadecimal(十六进制) is compact
  - A byte(字节) is composed of 8 bits
  - One byte(字节) can thus be expressed by 2 digits in hexadecimal(十六进制)
    - $(11101111)_2 \rightarrow EF_h$
- Simple to convert them to binary(二进制)

Conversions(转换) Between Power-of-2 Radices

- Because 16 = 2^4, a group of 4 bits is easily recognized as a Hexadecimal(十六进制) digit
- Convert a binary(二进制) number to Hex (十六进制) number



Conversions(转换) Between Number Systems

- Binary(二进制) to Decimal(十进制)

$(10010011)_2 = 1 * 2^7 + 1 * 2^4 + 1 * 2^1 + 1 * 2^0$

$= 128 + 16 + 2 + 1$

$= (147)_{10}$

- Hexadecimal(十六进制) to Decimal(十进制)

$21F_h = 2 * 16^2 + 1 * 16^1 + 15 * 16^0$

$= 512 + 16 + 15$

$= (543)_{10}$

Conversions(转换) Between Number Systems

Decimal to binary(二进制)

Quotient(商) Remainder(余数)

2) 28	(	0
2) 14	(	0
2) 7	(	1
2) 3	(	1
2) 1	(	1
0		

Decimal(十进制) to hexadecimal(六进制)

16) 28	(	C
16) 1	(	1
0		

Reading the remainders(余数) from bottom to top, we have

$(28)_{10}=(11100)_2$ $(28)_{10}=(1C)_{16}$

Basic Arithmetic(算术) Operations with Binary(二进制) Numbers

- Rules for Binary(二进制) Addition
  - 1+1=0, with one to carry to the next place

				1
0	0	1	1	+1
+0	+1	+0	+1	+1
0	1	1	10	11

Example

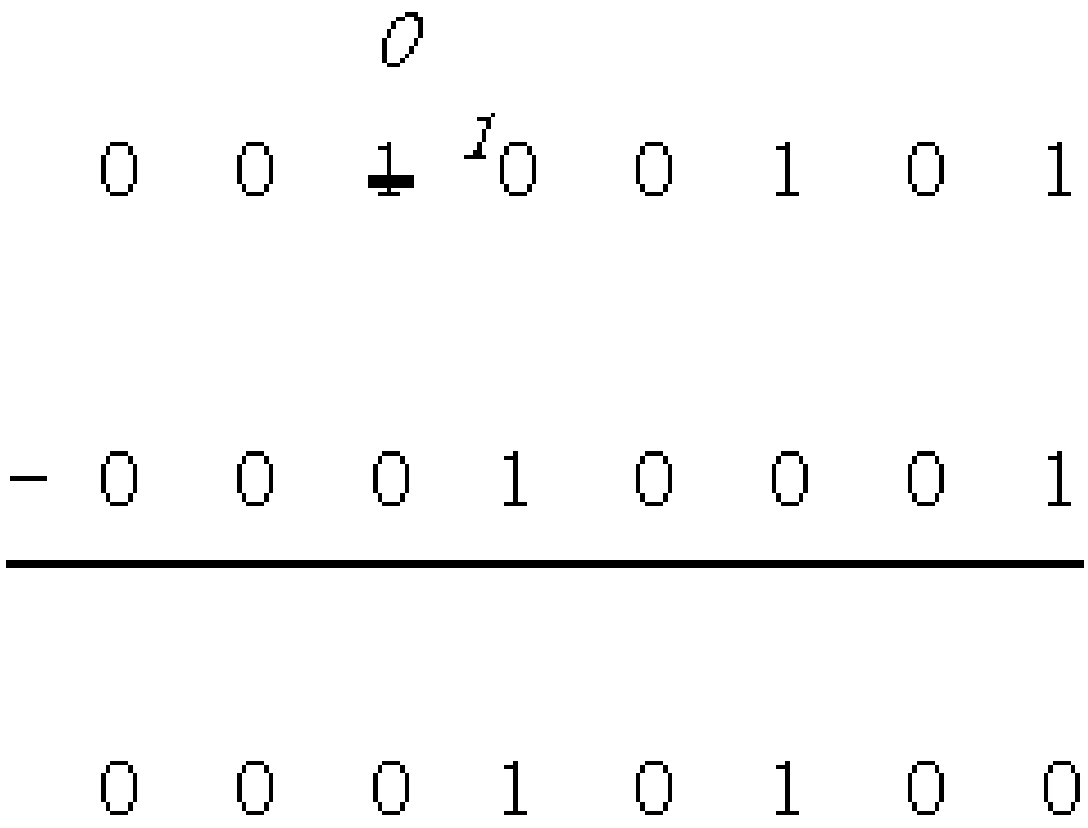
$$
\begin{array}{r}
\ {\scriptstyle 1\ 1\ 1\ \ \ \ } \\
\ 0\ 0\ 1\ 1\ 0 \\
+\ 0\ 1\ 1\ 1\ 1 \\
\hline
1\ 0\ 1\ 0\ 1
\end{array}
$$

Basic Arithmetic(算术) Operations with Binary(二进制) Numbers

- Rules for Binary(二进制) Subtraction(减法)
  - 1 - 0 = 1
  - 1 - 1 = 0
  - 0 - 0 = 0
  - 0 - 1 = 1 … borrow 1 from the next most significant bit

```
  51      minuend (被减数)
− 22      subtrahend (减数)
  29      difference.
```

Example


0
0 0 1 10 0 1 0 1
minuend(被减数)
- 0 0 0 1 0 0 0 1
subtrahend(减数)
0 0 0 1 0 1 0 0
difference

Binary(二进制) representation of Numbers

- How to know the binary(二进制) representation of a number (assume one byte(字节) is used to represent a number)
  - Positive number
    - MSB is 0
    - Directly convert the number to binary(二进制)
    - E.g., 28 -> 0001 1100
    - Range from 1 to 127
  - zero
    - 0000 0000

Binary(二进制) representation of Numbers

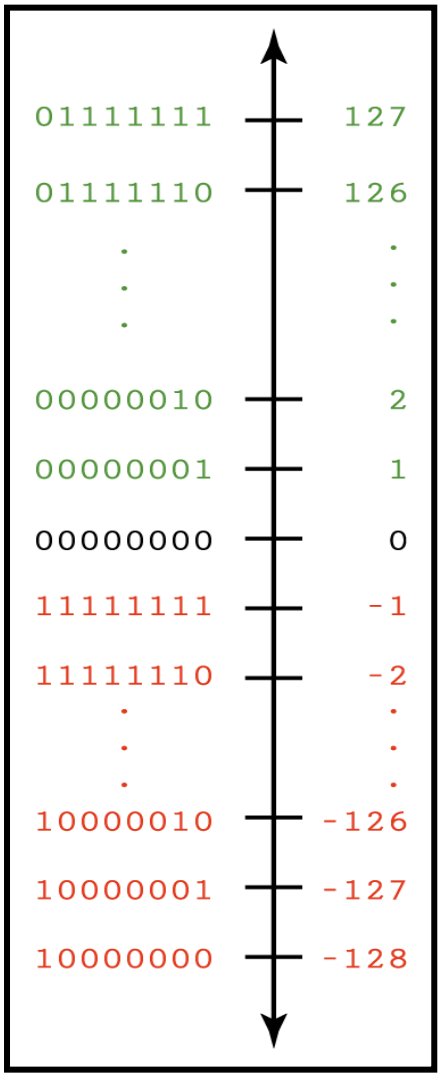
- Negative number
  - MSB is 1
  - Use 2's complement(补集) to get the representation of negative integers.
  - Three steps
    1. Convert the absolute value (绝对值) to binary(二进制)
    2. Invert the digits
    3. Add 0000 0001
- E.g., -28
  1. 28 -> 0001 1100 (absolute value to binary(二进制))
  2. 0001 1100 -> 1110 0011 (Invert the digits)
  3. 1110 0011 -> 1110 0100 (Add 0000 0001)

Overflow(溢出)

- Overflow(溢出) occurs when the value that we compute cannot fit into the number of bits we have allocated for the result

```
  1111111
+ 0000011
  10000010
```

```
  127
+   3
```



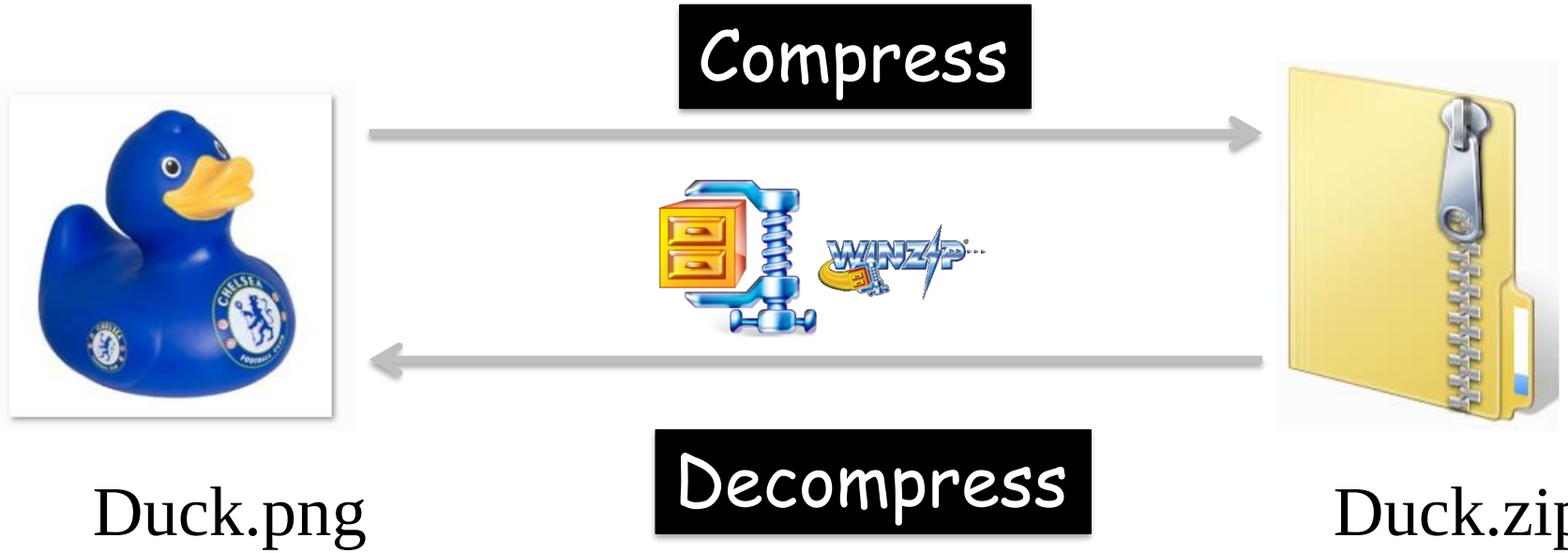
Data Compression(压缩)

- The goal is to represent the information as accurately as possible using the fewest number of bits.
- Data Compression(压缩) techniques can be
  - Lossless (数据不丢失的)
    - Decrease data size by optimizing the encoding
    - Does not lose any of the original information
  - Lossy (数据丢失的)
    - Non-crucial information may be given up

But sometimes the data just get really **BIG** …



Lossless Compression(压缩)



Compression software: 7-Zip, WinRAR

Lossy Compression(压缩)



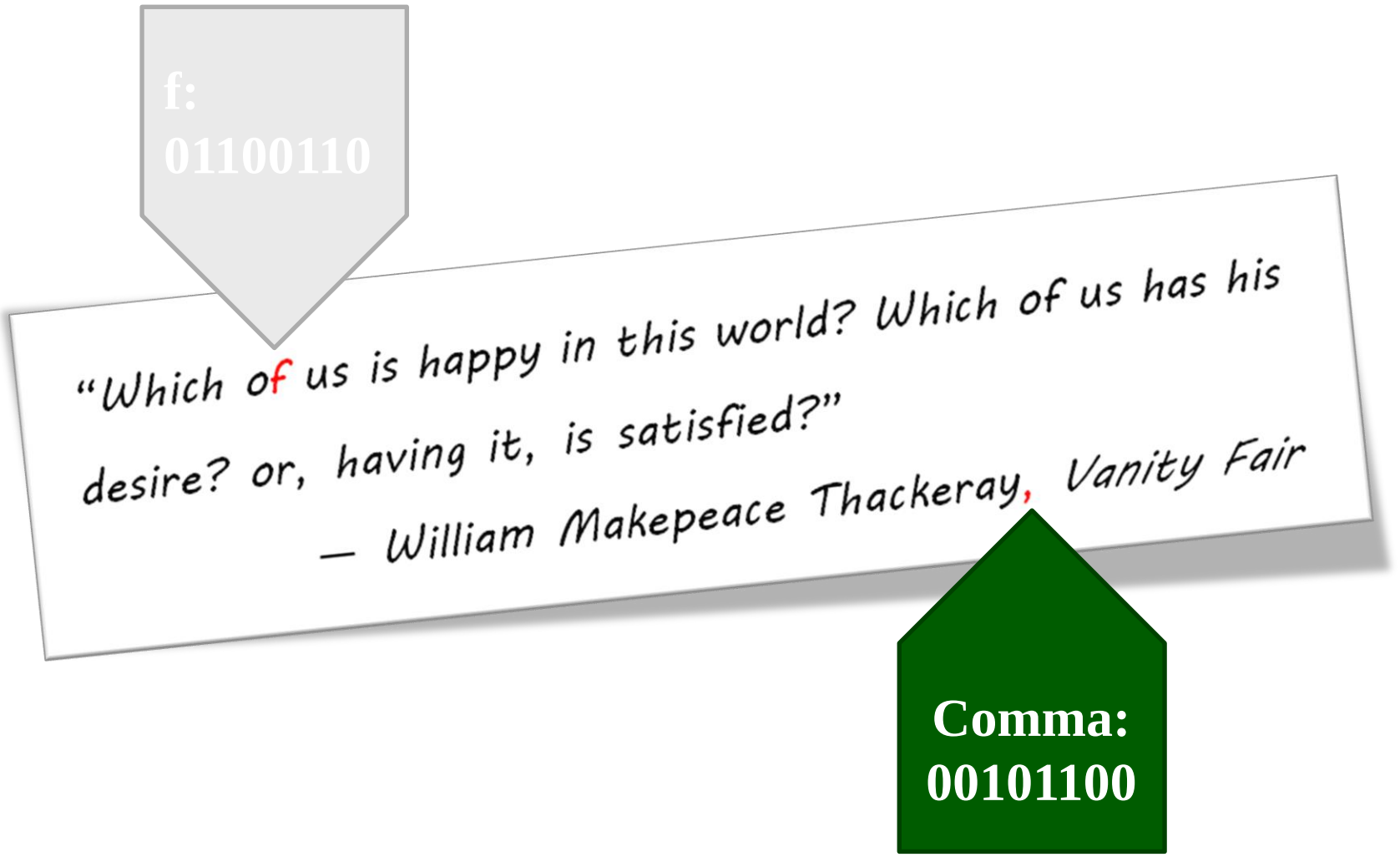
Duck.png

Duck.gif

Representing Text(文本)

- A text document can be decomposed into chapters, paragraphs, sentences,words, and ultimately, individual characters
- Represent every character(字符) that may appear
  - "a, b, c, d, … "
  - "1, 2, 3, 4, …"
  - "+, -, *, /, …"
  - "高, 天, 孤, 月, …"
- The general approach is to assign each character a binary(二进制) code
  - 'a' → $(01100001)_2$ → $(97)_{10}$ → 61_h

Representing Text (文本)



ASCII Character Set

- ASCII stands for American Standard Code for Information Interchange
  - Use 8 bits to represent
  - Designed for English
  - 0-31: for control characters, cannot be displayed
  - Uses 8 bits to represent a character
    - E.g.,
      - 'a': 97 ('b': 98 ..... inferred from the code of 'a')
      - 'A': 65 ('B': 66 ..... inferred from the code of 'A')
      - '0': 48 ('1': 49 ..... inferred from the code of '0')
  - It is better to remember ASCII for 'a', 'A', and '0'.

Basic ASCII

Char	Dec	Oct	Hex	Char	Dec	Oct	Hex	Char	Dec	Oct	Hex	Char	Dec	Oct	Hex
(nul)	0	0000	00	(sp)	32	0040	20	@	64	0100	40	`	96	0140	60
(soh)	1	0001	01	!	33	0041	21	A	65	0101	41	a	97	0141	61
(stx)	2	0002	02	"	34	0042	22	B	66	0102	42	b	98	0142	62
(etx)	3	0003	03	#	35	0043	23	C	67	0103	43	c	99	0143	63
(eot)	4	0004	04	$	36	0044	24	D	68	0104	44	d	100	0144	64
(enq)	5	0005	05	%	37	0045	25	E	69	0105	45	e	101	0145	65
(ack)	6	0006	06	&	38	0046	26	F	70	0106	46	f	102	0146	66
(bel)	7	0007	07	'	39	0047	27	G	71	0107	47	g	103	0147	67
(bs)	8	0010	08	(	40	0050	28	H	72	0110	48	h	104	0150	68
(ht)	9	0011	09	)	41	0051	29	I	73	0111	49	i	105	0151	69
(nl)	10	0012	0a	*	42	0052	2a	J	74	0112	4a	j	106	0152	6a
(vt)	11	0013	0b	+	43	0053	2b	K	75	0113	4b	k	107	0153	6b
(np)	12	0014	0c	,	44	0054	2c	L	76	0114	4c	l	108	0154	6c
(cr)	13	0015	0d	-	45	0055	2d	M	77	0115	4d	m	109	0155	6d
(so)	14	0016	0e	.	46	0056	2e	N	78	0116	4e	n	110	0156	6e
(si)	15	0017	0f	/	47	0057	2f	O	79	0117	4f	o	111	0157	6f
(dle)	16	0020	10	0	48	0060	30	P	80	0120	50	p	112	0160	70
(dc1)	17	0021	11	1	49	0061	31	Q	81	0121	51	q	113	0161	71
(dc2)	18	0022	12	2	50	0062	32	R	82	0122	52	r	114	0162	72
(dc3)	19	0023	13	3	51	0063	33	S	83	0123	53	s	115	0163	73
(dc4)	20	0024	14	4	52	0064	34	T	84	0124	54	t	116	0164	74
(nak)	21	0025	15	5	53	0065	35	U	85	0125	55	u	117	0165	75
(syn)	22	0026	16	6	54	0066	36	V	86	0126	56	v	118	0166	76
(etb)	23	0027	17	7	55	0067	37	W	87	0127	57	w	119	0167	77
(can)	24	0030	18	8	56	0070	38	X	88	0130	58	x	120	0170	78
(em)	25	0031	19	9	57	0071	39	Y	89	0131	59	y	121	0171	79
(sub)	26	0032	1a	:	58	0072	3a	Z	90	0132	5a	z	122	0172	7a
(esc)	27	0033	1b	;	59	0073	3b	[	91	0133	5b	{	123	0173	7b
(fs)	28	0034	1c	<	60	0074	3c	\	92	0134	5c	|	124	0174	7c
(gs)	29	0035	1d	=	61	0075	3d	]	93	0135	5d	}	125	0175	7d
(rs)	30	0036	1e	>	62	0076	3e	^	94	0136	5e	~	126	0176	7e
(us)	31	0037	1f	?	63	0077	3f	_	95	0137	5f	(del)	127	0177	7f

Unicode Character Set



- Offers multilingual (多种语言) support
- Uses 16 bits to represent a character
  - Question: How many characters can Unicode represent?
- Designed as a superset (父集) of ASCII
  - The first 256 characters in Unicode correspond exactly to those in ASCII
  - Makes it convenient to convert between ASCII and Unicode

Unicode

Code (Hex)	Character	Source
0041	A	English (Latin)
042F	Я	Russian (Cyrillic)
OE09	ก	Thai
13EA	Ꮺ	Cherokee
211E	℞	Letterlike Symbols
21CC	⇌	Arrows
282F	⠯	Braille
345F	㑯	Chinese/Japanese/ Korean (Common)

Code (Hex)	Description	Characters	Hex Values
Alphabets	Latin, Cyrillic, Greek, etc	8192	0000 - 1FFF
Symbols	Dingbats, Mathematical, etc	4096	2000 - 2FFF
CJK	CH, J., K. phonetic	4096	3000 - 3FFF
Han	Unified Chinese	40,960	4000 - DFFF
Han	Expansion	4096	E000 - EFFF
User Defined		4096	F000 - FFFF

Exercises

Please show your steps of conversion(转换) clearly.

- Convert 199_{10} to binary
- Convert 255_{10} to binary
- Convert 255_{10} to hexadecimal
- Convert 255_{8} to decimal
- Convert 123_{16} to decimal
- 00011010 + 00001100 = ?
- 00110011 - 00010110 = ?

Exercises

- What is the ASCII code for 'h'?
- What is the ASCII code for 'Z'
- What is the ASCII code for '5'
- What are the relationships between ASCII code of English letters?
- How many bits are used in ASCII code
- How many bits are used in Unicode
- A Unicode has 16 bits, so how many different characters can Unicode represent?